

LEGAL METROLOGY DETAILED TEST REPORT

NON-AUTOMATIC WEIGHING INSTRUMENT (NAWI) - TECHNICAL AUDIT RECORD

Regulatory Standard: OIML R 76-1:2006 | Evaluation Mode: TYPE_EVALUATION

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REPORT NO: R76-2026-000073

QR VERIFICATION ID: R76-2026-D6F438AF

1. LABORATORY INFORMATION

Facility: Central Legal Metrology Testing Laboratory

Address: Central Metrology Directorate

Assigned Technician: Pramod Patil

2. ENVIRONMENTAL CONDITIONS

Ambient Temperature: 21.8 °C

Relative Humidity: 48 %

Atmospheric Pressure: 1013.25 hPa

Compliance: Within standard metrological tolerance limits

3. INSTRUMENT TECHNICAL SPECIFICATION

Manufacturer: Mettler Toledo

Model: ICS425

Serial No: MT-55198

Type: Non-Automatic Weighing

Accuracy Class: Class III

Max: 15 kg

Min: 0.1 kg

Instrument (Compact Industrial Scale)

Verification (e): 0.005 kg

Resolution (n): 3,000 intervals

Unit: kg

Interval (d): 0.001 kg

Indication: Electronic Digital

4. DETAILED METROLOGICAL TEST AUDITS & CALCULATION TRACES

TEST: WEIGHING PERFORMANCE & INTRINSIC ERROR (WEIGHING PERFORMANCE) Reference: OIML R 76-1:2006 Clause 3.5.1 & A.4.5 STATUS: PASS

Formulae: $P = I + 0.5e - 9d$ & $w \text{ Error } E = P - L$ | Corrected Error $E_c = E - E_0$

No.	Phase	Load L (kg)	Ind. I (kg)	dL	E0	Raw E	Corr. Ec	MPE (+/-)	Decision
1	LOADING	0.000	0.000	0.001	0.002	0.002	0.000	+/-	PASS
2	LOADING	3.750	3.750	0.001	0.002	0.002	0.000	+/-	PASS
3	LOADING	7.500	7.501	0.001	0.002	0.003	0.001	+/-	PASS
4	LOADING	15.000	15.001	0.001	0.002	0.003	0.001	+/-	PASS

TEST: REPEATABILITY TEST (REPEATABILITY) Reference: OIML R 76-1:2006 Clause 3.6.1 & A.4.5 STATUS: PASS

Evaluation Formula: $\Delta I = I_{\max} - I_{\min}$ | Applicable Limit: MPE at Test Load

RECORDED TEST READINGS (RUNS):

R1: 7.500 kg | R2: 7.500 kg | R3: 7.500 kg | R4: 7.500 kg | R5: 7.500 kg | R6: 7.500 kg | R7: 7.500 kg | R8: 7.500 kg | R9: 7.500 kg | R10: 7.500 kg

Test Load: 7.500 kg

Max Indication (I_{max}): 7.500 kg

Min Indication (I_{min}): 7.500 kg

Difference (Delta I): 0.001 kg

Applicable MPE: 0.005 kg

Decision: PASS

TEST: ECCENTRIC LOADING TEST (ECCENTRIC_LOADING) Reference: OIML R 76-1:2006 Clause 3.6.2 & A.4.5 STATUS: PASS

Evaluation: Test Load = 1/3 Max (5.000 kg) | Error at each position <= MPE

Position	Applied Load (kg)	Indication (kg)	dL	Raw Error E	Corrected Ec	MPE	Result
CENTER	5.000	5.000	0.001	0.002	0.002	±0.005	PASS
FRONT_LEFT	5.000	5.001	0.001	0.003	0.003	±0.005	PASS
BACK_LEFT	5.000	5.001	0.001	0.003	0.003	±0.005	PASS
BACK_RIGHT	5.000	5.001	0.001	0.003	0.003	±0.005	PASS
FRONT_RIGHT	5.000	5.001	0.001	0.003	0.003	±0.005	PASS

TEST: ZERO-SETTING & ZERO-TRACKING EVALUATION (ZERO SETTING)

Reference: OIML R 76-1:2006 Clause 4.5 & A.4.2 STATUS: PASS

Initial Indication: 0.000 kg

Changeover 9D 0.003 kg

Calculated Zero Error E0: 0.000 kg

Applicable Limit: ±0.25 e (±0.001 kg)

Result: PASS

TEST: DISCRIMINATION TEST (DISCRIMINATION)

Reference: OIML R 76-1:2006 Clause 3.8 & A.4.8 STATUS: PASS

Requirement: Extra load of 1.4 d smoothly placed must step indication to I + d (Clause 3.8 / A.4.8)

Base Load (L): 9.000 kg

Initial Indication: 9.000 kg

Additional Load: 0.0014 kg (1.4 d)

Resulting Indication: 9.001 kg

Observed Shift (9D) 0.0010 kg

Mandated Shift: 0.0010 kg (1.0 d)

Decision: PASS

Note: Addition of 1.4d extra load (0.0014 kg) produced the mandated indication st

TEST: MULTIPLE INDICATING DEVICES COMPARISON (MULTIPLE INDICATING DEVICES)

Reference: OIML R 76-1:2006 Clause 3.6.3 STATUS: NOT_APPLICABLE

This test module is not applicable to the current instrument configuration.

TEST: DIFFERENT POSITIONS OF EQUILIBRIUM (DIFFERENT POSITIONS OF EQUILIBRIUM)

Reference: OIML R 76-1:2006 Clause 3.6.4 STATUS: NOT_APPLICABLE

This test module is not applicable to the current instrument configuration.

OVERALL VERIFICATION RESULT: PASSED (COMPLIANT)

All 5 applicable OIML R 76-1:2006 metrological tests have been verified compliant with Maximum Permissible Errors.

PUBLIC VERIFICATION RECORD & TRACEABILITY

Permanent Verification ID: R76-2026-D6F438AF

Authentic URL to inspect full measurement calculations:

<http://localhost:5173/public/report/R76-2026-D6F438AF>

OFFICIAL ATTESTATION & SIGN-OFF

Officer: Pramod Patil

Facility: Central Legal Metrology Testing Laboratory

[DIGITALLY RECORDED & ARCHIVED]

